

Multiple Choice

1. **Answer:** C

Options: A) Deficiency of leptin caused by a mutation in the leptin gene; B) Lack of leptin release from the adipose tissue; C) High concentrations of leptin without reduction in energy intake; D) Low concentrations of leptin with increased energy intake

Note: Correct option: C - High concentrations of leptin without reduction in energy intake

2. **Answer:** D

Options: A) Suppresses milk production; B) Alters the composition of breast milk; C) Increases the risk of early cessation of breastfeeding; D) all of the options given are correct

Note: Correct option: D - all of the options given are correct

3. **Answer:** C

Options: A) BMI; B) DXA; C) Four-compartment model; D) MRI

Note: Correct option: C - Four-compartment model

4. **Answer:** D

Options: A) fasting blood glucose $\leq$ 140 mg/dl; B) random glucose $>$ 160 mg/dl; C) 2 hour post prandial glucose $\geq$ 126 mg/dl; D) fasting blood glucose $\geq$ 126 mg/dl

Note: Correct option: D - fasting blood glucose $\geq$ 126 mg/dl

5. **Answer:** A

Options: A) Folate; B) Riboflavin; C) Thiamin; D) Vitamin C

Note: Correct option: A - Folate

True / False

6. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

7. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

8. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Deuterium'.

9. **Answer:** True

Options: A) True; B) False

Note: LLM-generated true statement based on MCQ.

10. **Answer:** False

Options: A) True; B) False

Note: LLM-generated false statement. Correct answer: 'Intake only of locally grown food in central Africa'.

Long Form Response

11. **Answer:** Alcohol. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

12. **Answer:** Vitamin B 6. Explain why this is the correct answer and provide supporting evidence. Reference key facts or concepts that support this choice.

Note: Long-form derived from MCQ; award credit for stating the correct idea and giving supporting reasoning.

End of Answer Key